# Technology Stack

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| **Date** | 08/07/2026 |
| **Team ID** | |
| **Project Name** | A comprehensive of measure of well being |
| **Maximum Marks** | 2 Marks |

## Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| **1** | **Frontend /**<br>Client-Side<br><br>(HTML5, CSS3,<br>JavaScript ES6+) | HTML5, CSS3,<br>Vanilla JavaScript | No build-tool overhead needed — the HDI computation engine is pure<br>client-side maths (geometric mean formula). HTML5 + CSS3 deliver a<br>fully responsive layout with interactive sliders, real-time score<br>display, and tier badge without any framework dependency. This keeps<br>the tool lightweight, fast-loading, and accessible on all devices. |
| **2** | **Backend /**<br>Server-Side<br><br>(No server required —<br>client-side only) | No backend<br>(Pure client-side<br>computation) | The UNDP HDI formula (geometric mean of three normalised sub-indices)<br>requires no server processing. All computation runs in the browser<br>using JavaScript. This eliminates server costs, removes latency, and<br>makes the tool instantly deployable as a static file on any CDN.<br>Future enhancement: optional Node.js API for bulk country processing. |
| **3** | **Database /**<br>Data Storage<br><br>(Browser localStorage<br>/ JSON) | Browser<br>localStorage<br>+ JSON config | HDI indicator goalposts (min/max bounds per UNDP) are stored as a<br>JSON configuration object in the JavaScript file. User session data<br>(last-used indicator values, selected scenario) is persisted via<br>browser localStorage. No external database is required since the tool<br>operates entirely on user-supplied inputs rather than stored records. |
| **4** | **Cloud / Hosting /**<br>Deployment | Vercel /<br>Netlify /<br>GitHub Pages | As a static HTML/CSS/JS application, the HDI Predictor deploys to |

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| | (e.g., Vercel,<br>Netlify, GitHub Pages) | | any static hosting platform in seconds with zero configuration.<br>Vercel and Netlify provide free tier CDN-backed hosting with HTTPS,<br>automatic deployment on git push, and global edge distribution —<br>ensuring fast load times for government and research users worldwide. |
| **5** | **Version Control**<br>& CI/CD<br><br>(GitHub +<br>GitHub Actions) | GitHub +<br>GitHub Actions | GitHub provides version-controlled source management with branching<br>for feature development (e.g. multi-country comparison, export PDF).<br>GitHub Actions automates deployment: on every push to main, the<br>static site is validated and deployed to the hosting platform.<br>This ensures a reproducible, reviewable, and auditable codebase. |
| **6** | **Third-Party APIs /**<br>Other Tools<br><br>(UNDP Data API,<br>jsPDF) | UNDP HDR API<br>+ jsPDF library | UNDP Human Development Report API (hdr.undp.org) provides official<br>published HDI scores for validation and country benchmarking. jsPDF<br>(open-source JS library) enables one-click export of the computed<br>HDI score, tier classification, and sub-index breakdown as a<br>formatted PDF report suitable for ministry or academic submission. |